Effects of spectral smearing on speech understanding and masking release in simulated bilateral cochlear implants

18 July 2022

[NOTE: Some of the figures are redundant, so we can choose which ones we'd actually like to include in the paper.]

1 Results

We first report descriptive statistics and multilevel modeling on the SRTs in each listening condition followed by the SRM results. All results and modeling were conducted in the RStudio computing environment (v. 2022.07.0) using the lme4 and lmerTest packages for modeling and ggplot2 for graphics (Bates et al., 2015; Kuznetsova, et al., 2017; Wickham, 2016).

1.1 SRT results

Unsurprisingly, SRTs were higher for diotic than dichotic masker presentation conditions (Figure 1 and Table 1). SRTs were also higher for the binaural broad channel interaction condition (L12R12) than the monaural broad-limited (L48R12) or binaural limited interaction conditions (L48R48) (Figure 2), suggesting a negative impact of increased channel interaction upon SRTs. The beneficial effect of the dichotic masker condition was consistent across the channel interaction conditions, but especially when the channel interaction was similar across ears (L12R12: 6.45 dB difference; L48R48: 6.51 dB).

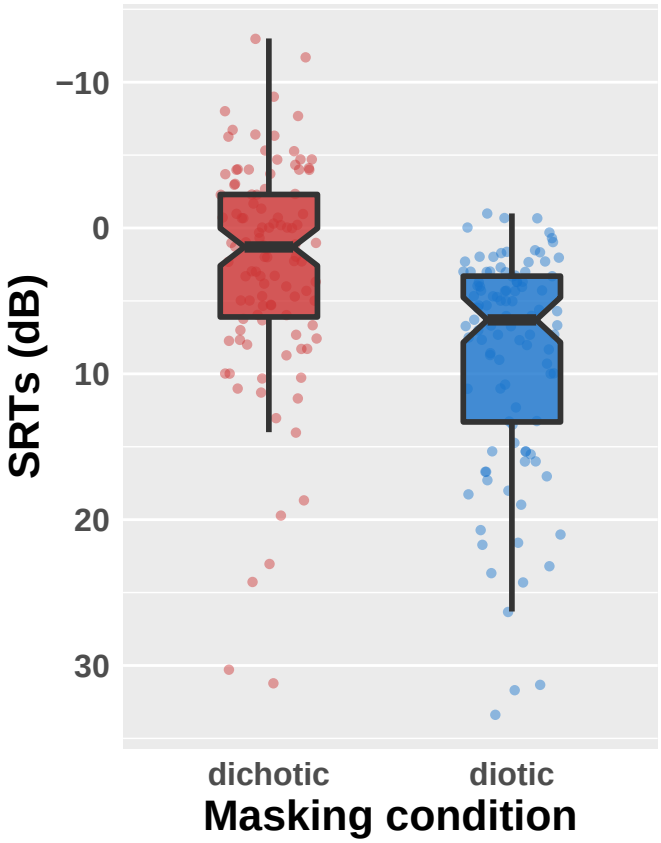


Figure 1: Speech recognition thresholds (SRTs) by listening condition. Smaller SRTs indicate better recognition of test material (i.e. at more challenging signal to noise ratios). Colored points represent individual listener responses; points are horizontally jittered for readability. Boxes indicate interquartile range and whiskers 1.5 x the interquartile range in each direction. Black horizontal lines indicate the median and notches 95% confidence intervals surrounding the median.

Table 1: Descriptive statistics of SRTs (dB) by listening condition: Mean (SD) range.

Masker presentation	Combined channel interaction conditions	Binaural broad (L12R12)	L limited, R broad (L48R12)	Binaural limited (L4
dichotic	2.65(7.69)-13–31.2	9.84(7.76)0–31.2	0.92(4.03)-8–8.7	-2.96(3.88)-13–5.3
diotic	8.8(7.6)-1–33.4	16.29(7.08)4.3–33.4	4.82(2.87)-0.7–11	3.55(2.16)-1–7.7

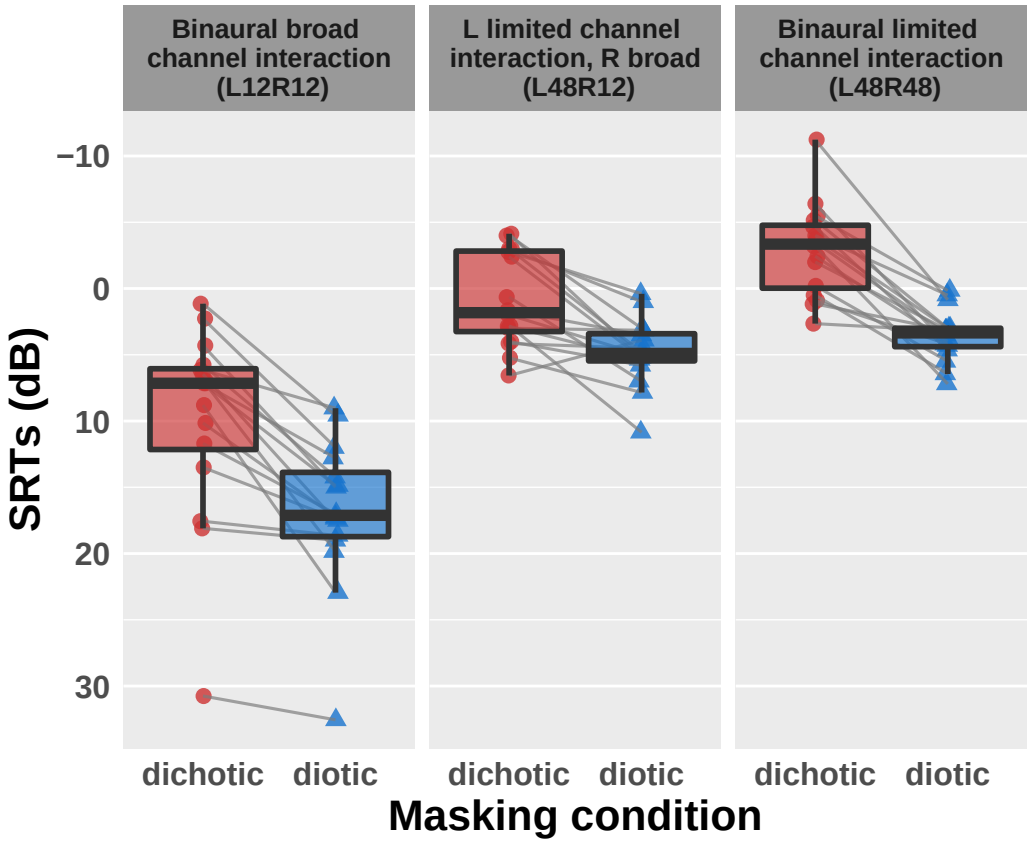


Figure 2: SRTs by channel interaction condition. Colored points represent individual listener responses; gray lines connect listener-level SRTs across masking conditions. Boxes indicate interquartile range and whiskers 1.5 x the interquartile range in each direction. Black horizontal lines indicate the median.

To complement description of these raw data, we additionally fit a linear mixed effects model to predict the effect of listening condition upon SRTs. The significance of potential model parameters was determined via comparing model log-likelihood values and AIC estimates. Categorical predictor variables were contrast-coded to assist in model interpretation. Pairwise comparisons were calculated using the emmeans package (Lenth, 2021). To account for individual variability and correlation stemming from participant-level effects, we included random intercepts by participant; all fixed effects were added step-wise to this baseline, random effect-only model.

Best model fit included the fixed effects of masker presentation (diotic vs. dichotic) and channel interaction (binaural broad vs. monaural limited and broad vs. binaural limited), but not their interaction (see Table 2 for summary). The lack of an interaction suggests that the benefit of dichotic > diotic is consistent in all three channel interaction conditions. Interpretation of model coefficients indicates a SRT increase of 5.7 dB in the diotic over dichotic conditions ($t=9.45$, $p<.001$). The model likewise suggests a 2.64 dB greater SRT in the monaural limited-broad channel interaction over binaural limited channel interaction conditions ($t=3.52$, $p=.002$) and a heightened SRT of 10.23 dB in the binaural broad channel interaction condition over the monaural limited-broad interaction condition ($t=14.01$, $p<.001$). See model summary for full pairwise comparisons. The significant effect of masker presentation, without an interaction, indicates SRM in all channel interaction conditions which will be explored in the following section.

1.2 SRM results

To further evaluate the effect of masker presentation, we evaluated a second outcome variable: SRM, or the difference between the average SRT in the diotic and dichotic masker conditions. Here, a higher value

Table 2: Model predicting the effect of listening condition upon SRTs.

	Estimate	S.E.	t.value	p-value
Intercept	10.33	0.93	11.12	<.001
Masker				
Diotic	5.70	0.60	9.45	<.001
Channel Interaction				
L12R12–L48R12	10.23	0.73	14.01	<.001
L12R12–L48R48	12.87	0.74	17.47	<.001
L48R12–L48R48	2.64	0.75	3.52	.002

indicates greater release from masking or the listener benefiting more from the dichotic listening condition during the task.

As previously mentioned, significant SRM was found for all channel interaction conditions. However, the SRM was greatest in the binaural broad and binaural limited channel interaction conditions (L12R12: M=6.98 (SD=4.16); L48R48: M=6.48 (SD=3.15), indicating greater SRM for same than different channel interaction (Figure 3).

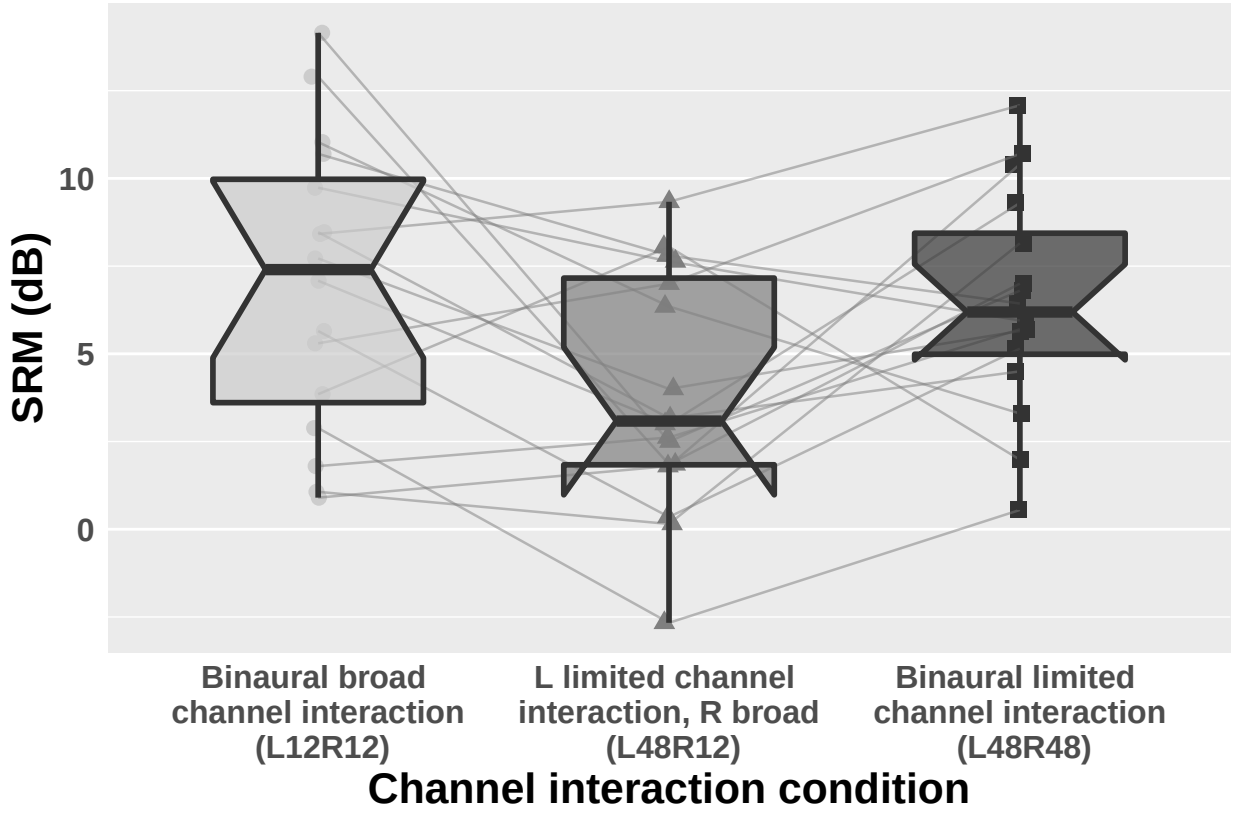


Figure 3: SRM (diotic-dichotic) by channel interaction condition. Higher values indicate greater benefit from dichotic over diotic listening conditions. Points represent individual listener responses; gray lines connect listener-level SRM values across channel interaction conditions. Boxes indicate interquartile range and whiskers 1.5 x the interquartile range in each direction. Black horizontal lines indicate the median and notches 95% confidence intervals surrounding the median.

To further evaluate these results, we fit another linear mixed effects model, controlling for participant-level effects in the random effect structure (random intercepts), to predict the effect of channel interaction condition upon SRM (see Table 3 for model summary and pairwise comparisons). Channel interaction condition improved upon a baseline, random effects-only model. Model coefficients show a 3.04 dB SRM benefit for the binaural broad over the broad-limited condition, but just a 0.5 dB SRM benefit for the binaural broad over binaural limited condition and -2.54 dB benefit for mixed channel interaction over limited interaction. Together, these results suggest that the most SRM occurs during *same* channel interaction conditions (broad or limited interaction), and that manipulating the degree of channel interaction affects SRM less.

Table 3: Model predicting the effect of channel interaction condition upon SRM.

	Estimate	S.E.	t.value	p value
Channel Interaction				
L12R12 - L48R12	3.04	0.71	4.31	<.001
L12R12 - L48R48	0.50	0.71	0.71	0.759
L48R12 - L48R48	-2.54	0.71	-3.60	0.002

2 Extra Figures

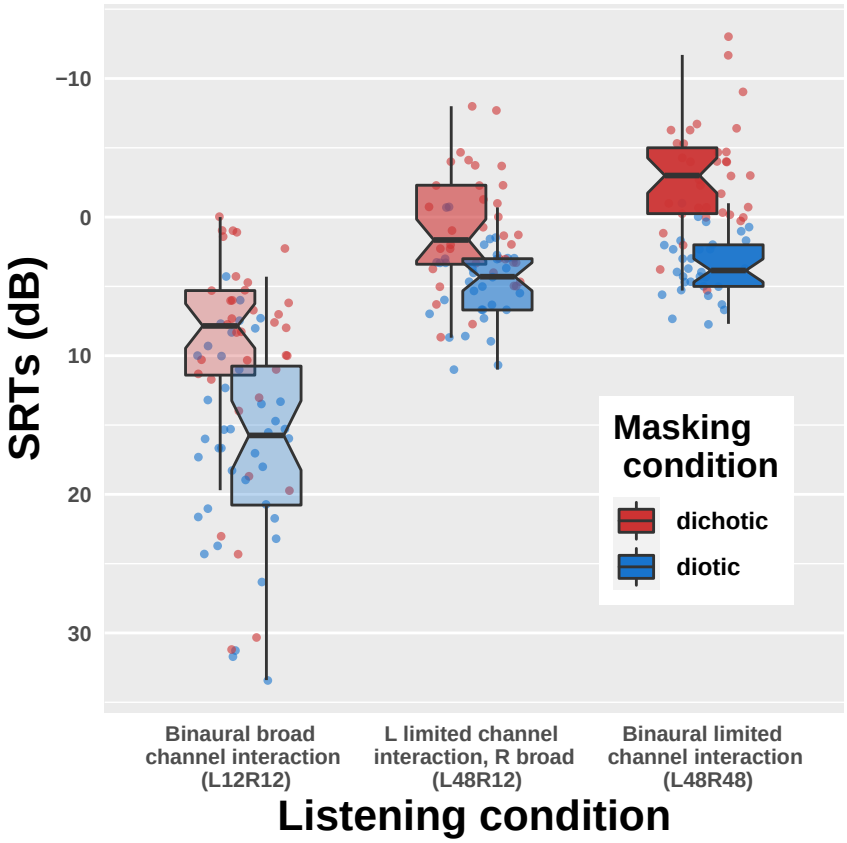


Figure 4: Speech recognition thresholds by listening condition and degree of channel interaction. Colored points represent individual listener responses; points are horizontally jittered for readability. Boxes indicate interquartile range and whiskers 1.5 x the interquartile range in each direction. Black horizontal lines indicate the median and notches 95% confidence intervals surrounding the median.